RiskPulseAI

Event-Driven Portfolio Stress Testing using Financial NLP

From Unstructured Financial Signals to Actionable Portfolio Risk Insights



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From Financial Noise to Portfolio Risk



Markets generate massive unstructured news and social data

Critical risk signals are difficult to identify, classify, and quantify quickly.

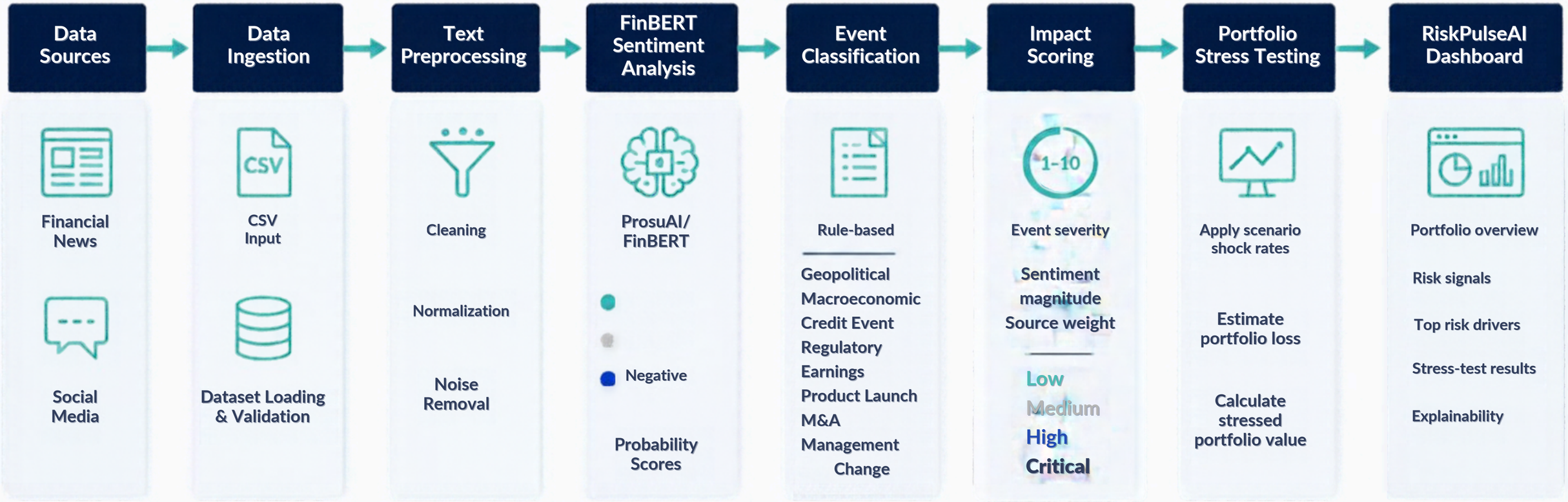
Portfolio managers need clear links between events and exposure

We turn noise into actionable insights for smarter stress testing _



Uses synthetic/public-style financial data for demonstration

RiskPulseAI System Architecture



Key Data Files

- news_data.csv
- social_data.csv
- portfolio.csv
- risk Signals.csv
- stress_test_results.csv

How the Risk Engine Works

From unstructured news to quantified portfolio stress

01 FINANCIAL NLP

Python, Hugging Face Transformers

ProsuAI/FinBERT model for sentiment probability scores



02 EVENT DETECTION

Rule-based keyword classification

Identifies financial event categories

Stores matched keywords as evidence

- earnings miss
- credit downgrade
- merger announced
- management change
- regulatory investigation

03 IMPACT SCORE

Impact = (Event Severity x 70%) + (Sentiment Severity x 30%), adjusted by Source Weight

Normalized to 1–10 scale

- Event severity — 70%
- Sentiment magnitude — 30%
- Source reliability weight
- Final score: 1–10
- Risk levels: Low / Medium / High / Critical

Low | Medium | High | Critical

04 PORTFOLIO STRESS ENGINE

Maps risk levels to scenario shock rates

Low	2%
Medium	5%
High	10%
Critical	20%

Calculates estimated loss and stressed portfolio value

Technology stack:
Python, Flask, Pandas, NumPy, Hugging Face Transformers, FinBERT, HTML, CSS, JavaScript, Pytest

Key Results – Portfolio Stress Scenario

Portfolio Value	Estimated Stress Loss	Stressed Portfolio Value	Portfolio Loss
$9.5M	$1.2M	$8.3M	12.63%

20 total risk signals analyzed

1 Critical event

12 High-risk events

Top risk drivers	Impact		Triggers
Alpha Motors – Geopolitical	10	Critical	supply chain disruption / geopolitical tensions
Beta Energy – Regulatory	8	High	regulatory investigation
Delta Bank – Credit Event	8	High	rising loan defaults
Epsilon Pharma – Regulatory	8	High	regulatory approval
Gamma Tech – Earnings	7	High	revenue forecast reduction

Stress losses are scenario-based estimates produced using predefined shock assumptions; they are not market forecasts.

From Risk Signals to Actionable Insights



DOMAIN IMPACT

Converts unstructured financial info into structured risk signals
Links events to portfolio holdings
Provides transparent risk drivers and evidence
Enables scenario-based portfolio stress analysis
Interactive dashboard for faster risk assessment

LIMTATIONS

Synthetic demonstration data

Rule-based event classification
Predefined shock rates as scenario assumptions

FinBERT sentiment may misclassify context-dependent language

FUTURE IMPROVEMENTS

Real-time financial news ingestion
More advanced event detection
Historical backtesting
Dynamic shock calibration
Time-series risk forecasting
Larger multi-asset portfolios

> “From unstructured financial data to explainable portfolio risk insights.”